

Track-System – Glossary

Last updated: 10 Oct 2026 (after step 4) | The words that keep coming back in this project, each one in plain language | **[NEW]** = added in step 4

1. The big picture – who talks to whom

FRONTEND (in the browser)	BACKEND (on the computer)	DATABASE
React + Vite	ASP.NET Core (C#)	SQL Server
--- axios sends HTTP req --->	endpoint /api/...	
(+ the token, [NEW])	--- EF Core: C# to SQL -->	
<----- JSON answer -----	<----- rows -----	

The frontend NEVER talks to the database directly. Always through the backend.

2. The platform words

Word	What it means / where we meet it
.NET	Microsoft's platform for running C# programs, like Node.js is for JavaScript. It includes the compiler, the libraries and the runtime. (dotnet --version)
ASP.NET Core	The part of .NET for building web servers and APIs. Our whole backend.
SDK / Runtime	SDK = the tools to build code. Runtime = only what is needed to run it. A developer needs the SDK. We installed the SDK.
Kestrel	The actual web server inside ASP.NET Core. It listens on the port and receives the requests. (The line "Now listening on: ...")
NuGet	The library store for .NET. Every PackageReference in the .csproj is downloaded from there. [NEW] dotnet add package <name> downloads one and adds it to the .csproj.
npm	The same idea for JavaScript. The frontend's package.json is its shopping list. (npm install)
Vite	The tool that runs the React project while developing and builds it for production. Usually port 5173. (npm run dev)
[NEW] Configuration	Everything .NET reads at startup, merged into one place: appsettings.json + User Secrets. The code reads it with builder.Configuration["Jwt:Issuer"].
[NEW] Section:Key (:)	The colon means "inside". Jwt:Key = the Key inside the Jwt section, whether it came from appsettings.json or from User Secrets.
[NEW] Development environment	The mode set in launchSettings.json (ASPNETCORE_ENVIRONMENT). User Secrets are loaded only in Development.

3. The web words

Word	What it means
HTTP request	A message the browser sends to a server: a method (GET/POST/...), an address, optional headers and optional body data.
GET / POST / PUT / DELETE	GET = give me data. POST = here is new data (for example login details). PUT = update something. DELETE = remove it.
Endpoint	One address on the server plus the code that answers it. Example: /api/health.
API	The whole collection of endpoints: the menu of everything the backend can do for the frontend.
REST	A common style for designing an API: addresses are nouns (/api/employees), methods are verbs.
JSON	The text format both sides speak, for example {"status":"ok"}.
Status code	A number describing the result: 200 OK, 201 created, 400 bad request, 401 not logged in, 403 not allowed, 404 address not found, 500 server crashed.

Word	What it means
[NEW] 405 Method Not Allowed	The address exists but not for this method. Opening a POST-only endpoint in the browser (which sends GET) gives a blank page with 405.
[NEW] 423 Locked	Our answer when the account is locked after 5 wrong passwords, so React can tell it apart from a wrong password (401).
[NEW] 429 Too Many Requests	The answer rate limiting will give when one IP sends too many attempts (planned).
[NEW] Header	A name: value line sent with a request or answer, outside the body. Example: Content-Type: application/json.
[NEW] Authorization header	The standard header for credentials. We send: Authorization: Bearer <token>.
[NEW] Bearer	"The bearer (holder) of this token". Whoever presents a valid token is treated as its owner, which is why a stolen token is dangerous until it expires.
[NEW] WWW-Authenticate	A header the server adds to a 401 explaining why the token was refused, e.g. error="invalid_token", "The signature is invalid" or "The token expired".
[NEW] Request body	The data part of a POST, for example {"eNum":1,"password":"..."}.
[NEW] IP address / ::1	The caller's network address, saved in LoginLogs. ::1 is localhost in IPv6 (the same as 127.0.0.1).
localhost	"This computer". Not reachable from outside, which is what we want while developing.
Port	The door number on the machine. 5163 = our API, 5173 = React.
axios	A JavaScript library, already installed in the frontend, that sends HTTP requests from React.
CORS	A browser security rule: a page from port 5173 may not call port 5163 unless the server allows it. Still to do.
Connection refused	The browser's "can't reach this page": nothing is listening on that address. The server is not running or the port is wrong.
camelCase JSON	ASP.NET Core writes C# names (FirstName) into JSON as firstName automatically.
[NEW] Rate limiting	Limiting how many requests one IP may send per minute. Planned for the login, against bots.

4. The database words

Word	What it means
SQL Server	Microsoft's database program. SQL Express is the free local edition.
SSMS	SQL Server Management Studio, the app where you look at the tables by hand. [NEW] F5 runs everything in the current tab; highlight lines first to run only them.
Connection string	One line saying which server, which database and how to log in. In appsettings.json.
Windows Authentication	Logging into the DB with your Windows user, so no password is written anywhere.
ORM	Object Relational Mapper: maps database rows to objects in code.
EF Core	Entity Framework Core, Microsoft's ORM. Translates our C# into SQL and rows back into objects.
DbContext	The EF Core class that represents the database. Every query goes through our AppDbContext.
DbSet	One line inside the DbContext representing one table. Ours: Employees and [NEW] LoginLogs.
Model / Entity	A plain C# class whose properties mirror the columns of one table (Employee.cs, [NEW] LoginLog.cs). EF Core fills it.
Migration	EF Core creating tables from C# classes. We do NOT use it: our tables already exist.
Provider	The package that knows one database's dialect. Ours: Microsoft.EntityFrameworkCore.SqlServer.
Primary key	The column that identifies each row uniquely (Employees.eNum, [NEW] LoginLogs.logID).

Word	What it means
IDENTITY	The DB fills the number automatically (1, 2, 3...). Numbers are never reused, even after a DELETE.
NULL / NOT NULL	NULL = the column may be empty. NOT NULL = a value is required.
DEFAULT	A value the DB fills in when none is given, for example role = 'Employee'.
Foreign key	A rule that a column must point to an existing row in another table (Employees.teamNum -> Teams). A JOIN does not need it; it only enforces the link.
[NEW] Log table without FK	LoginLogs.eNum has no foreign key on purpose: it stores the number that was typed, so attempts by unknown numbers are logged too and history survives a deletion.
[NEW] sys.foreign_keys	A system view listing every foreign key, used to find an auto-generated FK name.
[NEW] DROP CONSTRAINT	ALTER TABLE ... DROP CONSTRAINT [name] removes a rule (like an FK) without touching the data.
[NEW] UPDATE ... SET ... WHERE	Changes existing rows. Always with WHERE, or every row changes.
[NEW] UTC	World time with no time zone and no daylight saving. Israel is UTC+3 in summer, UTC+2 in winter. lockoutEnd is stored in UTC.
[NEW] GETDATE() / GETUTCDATE()	SQL Server's local time / UTC time. Compare lockoutEnd with GETUTCDATE().
[NEW] AT TIME ZONE	Converts a time for display: lockoutEnd AT TIME ZONE 'UTC' AT TIME ZONE 'Israel Standard Time'.
Empty result []	The query worked but the table has no rows.

5. The C# and code words

Word	What it means
Namespace	The address of a class so other files can find it with using Ours: TrackSystem.Api.Data, .Models, [NEW] .Dtos.
[NEW] using	A line at the top that lets the file see the classes of a namespace, e.g. using System.Security.Claims.
Class / Property	A class is a blueprint. Its properties are the fields inside it.
Inheritance (:)	class AppDbContext : DbContext receives all of DbContext's abilities.
Lambda (=>)	A small function written inline and run later, e.g. the code inside MapGet.
async / await	Waiting for the DB without freezing the server.
Dependency Injection	You ask for something in the parentheses and the server hands you a ready one (AppDbContext db, [NEW] HttpContext http).
Service registration	Teaching the server how to build those objects (builder.Services...), before Build().
Middleware	Code every request passes through before the endpoint. [NEW] Ours: UseAuthentication and UseAuthorization.
Minimal API	Endpoints written directly in Program.cs (the older style is Controllers).
Attribute []	A label in square brackets about the ONE thing right after it. Example: [Key].
[Key]	Tells EF Core which property is the primary key (EEnum, [NEW] LogId).
EEnum vs enum	EEnum is our property name; enum is a C# keyword. C# is case-sensitive, so they never clash.
Nullable ?	int? or string? means "may be empty": the C# side of a NULL column.
= string.Empty	A starting value of "" so a required text property is never null.
Types mapping	nvarchar -> string, int -> int, bit -> bool, datetime -> DateTime.
LINQ	C#'s query methods (Select, Where...). EF Core turns them into SQL.

Word	What it means
.Select(...)	Chooses which fields to take, which is how the password never leaves the DB.
Anonymous object	new { ... } with no class name, used to shape the JSON output.
ToListAsync()	Runs the query and returns the rows as a list.
[NEW] FirstOrDefaultAsync(...)	Returns the first matching row, or null if none. Used to find the employee by eNum.
Results.Ok(...)	Returns HTTP 200 with the object as JSON.
[NEW] Results.Json(obj, statusCode)	Returns any status code with a JSON body. Used for 401 and 423 with a message.
[NEW] Results.Text(...)	Returns plain text (used once, in the temporary hash endpoint).
[NEW] StatusCodes.Status401 Unauthorized	Built-in named constants for the status numbers (401, 423...).
DTO	Data Transfer Object: a class describing exactly what goes in or out of an endpoint. [NEW] First one in use: Dtos/LoginRequest.cs.
[NEW] record	A short way to write a class that only holds data: public record LoginRequest(int ENum, string Password);
[NEW] Change tracking	EF Core remembers the original values of loaded rows. Changing a property marks the row as changed.
[NEW] Add(...)	db.LoginLogs.Add(row) puts a new row on EF's "to save" list. Nothing is written yet.
[NEW] SaveChangesAsync()	Writes everything on the "to save" list to SQL Server in one trip (UPDATEs and INSERTs together).
[NEW] Local function	A small function written inside another one. AddLog(...) inside the login endpoint.
[NEW] HttpContext	The object describing the current request: headers, caller's IP (http.Connection.RemoteIpAddress), user.
[NEW] ?? (null-coalescing)	"If the left side is null, use the right side." Used to stop the server when Jwt:Key is missing.
[NEW] ?. (null-conditional)	"Only continue if not null." RemoteIpAddress?.ToString() gives null instead of crashing.
[NEW] DateTime.UtcNow	The current time in UTC. Used for lockoutEnd and token expiry.
[NEW] TimeZoneInfo	Converts UTC to a time zone. "Asia/Jerusalem" gives Israel time for LoginLogs.
[NEW] GetValue<int>(...)	Reads a setting and converts it to a number (ExpiresMinutes).
[NEW] Method chaining	} followed by .RequireAuthorization(); calls another method on the result, so the statement ends only after the last call.

6. The security words

Word	What it means
Hashing	Turning a password into scrambled text that cannot be turned back. We store and compare hashes only.
BCrypt	The hashing library we use. Deliberately slow, which makes guessing attacks expensive.
[NEW] HashPassword / Verify	BCrypt.HashPassword(pwd) makes a 60-char hash. BCrypt.Verify(typed, hash) returns true/false by re-hashing the typed password with the salt stored in the hash.
[NEW] \$2a\$11\$...	The hash format: version (\$2a\$), work factor (11 = 2 ¹¹ rounds), 22 chars of salt, then the hash.
[NEW] Work factor (cost)	How many rounds BCrypt runs. Higher = slower = harder to crack.
Salt	Random characters added before hashing, so the same password gives different hashes.
[NEW] Lockout	After 5 wrong passwords the account is locked for 15 minutes (failedLoginAttempts, lockoutEnd). A correct login resets the counter.

Word	What it means
[NEW] Generic error message	The same 401 text for a wrong number and a wrong password, so nobody learns which employee numbers exist (user enumeration).
[NEW] Timing attack	Learning something from how long an answer takes. Unknown eNums answer faster today; a dummy BCrypt check is planned.
[NEW] Audit log	A record of events for later checks. LoginLogs: every attempt, with Israel time, IP and result.
Authentication	Who are you? The login, and the token check on every request.
Authorization	What are you allowed to do? The rules: which endpoint needs a token or a role.
JWT	JSON Web Token: a signed string the server gives after a successful login. React sends it back on every later request.
[NEW] Header . Payload . Signature	The 3 parts of a JWT, separated by dots. Header = algorithm (HS256). Payload = the claims (readable by anyone). Signature = proof it was made with our key and not changed.
[NEW] Claim	One piece of data inside the token: a name and a text value.
[NEW] sub	Subject: who the token belongs to. Ours: the eNum, e.g. "1".
[NEW] role	What the user is (Employee / Manager / Admin). Used later for role rules.
[NEW] mustChangePassword	Our own claim: "true" on first login. Will limit that token to the change-password page.
[NEW] iss / aud	Issuer = who made the token (TrackSystem.Api). Audience = who it is for (TrackSystem.Client).
[NEW] exp / iat / nbf	Expires at / issued at / not valid before, as seconds since 1/1/1970 (UTC).
Token expiry	Our tokens live 60 minutes from login (Jwt:ExpiresMinutes). After that: 401 and log in again.
[NEW] Signing key	The secret (64 random bytes, in User Secrets) that signs and checks tokens. Whoever has it can forge tokens.
[NEW] Symmetric key / HMAC-SHA256 (HS256)	Symmetric = the same key signs and checks. HS256 = the signing method: a SHA-256 hash mixed with the key.
[NEW] SecurityTokenDescriptor	The "recipe" for one token: claims, issuer, audience, expiry, how to sign.
[NEW] JsonWebTokenHandler	The "printer" that turns the recipe into the header.payload.signature text.
[NEW] Stateless	The server keeps no list of tokens. Each request carries its token and is checked on the spot. Logout = React deletes its copy.
[NEW] AddAuthentication / AddJwtBearer	Registers the token guard and how it checks tokens.
[NEW] TokenValidationParameters	The guard's checklist: signature, issuer, audience, lifetime, plus which claims are the name (sub) and the role.
[NEW] ClockSkew	Allowed clock difference when checking expiry. Default 5 minutes; ours is 0, so 60 minutes means exactly 60.
[NEW] MapInboundClaims = false	Stops .NET from renaming sub and role to old long URL names (a compatibility habit from older Microsoft login systems).
[NEW] UseAuthentication	Middleware: reads the Bearer token on every request and identifies the caller. Blocks nothing by itself.
[NEW] AddAuthorization / UseAuthorization	Registers and runs the rules system. Blocks only endpoints that have a rule.
[NEW] RequireAuthorization()	The rule on one endpoint: no valid token -> 401, the code never runs.
User Secrets	Secret settings stored on your own PC, outside the project folder (AppData\Roaming\Microsoft\UserSecrets\<id>\secrets.json).
[NEW] UserSecretsId	The name of the project's secrets folder, written in the .csproj. Not secret.
Never expose the hash	Even a hashed password is never sent to the browser.

Word	What it means
[NEW] jwt.io	A website that decodes a token so you can read its payload. Paste test tokens only, never the key.

7. Tools and workflow

Word	What it means
git add / commit / push	add = choose the changes. commit = save a snapshot with a message. push = upload to GitHub.
[NEW] git status	Shows what changed before you commit. Check it every time: no secrets, no *.http files.
.gitignore	The list of files git must ignore: bin, obj, node_modules, secrets, [NEW] *.http.
bin / obj	Folders .NET generates while building. Never edit them, never commit them.
Build vs Run	Build = compile. Run = start. dotnet run does both; dotnet build only checks it compiles.
Hot reload	The backend does not reload by itself: after every change, Ctrl+C and dotnet run again.
Working directory	The folder the terminal is in. dotnet commands run in Backend\TrackSystem.Api.
cd	Change directory, e.g. cd Backend\TrackSystem.Api.
MSB1003	dotnet found no .csproj in the current folder. Fix: cd into the project folder.
Script Table as	SSMS menu that writes the SQL of an existing table. [NEW] Do not press F5 in that tab: it tries to create the table again ("There is already an object named...").
[NEW] dotnet user-secrets init / set / list	init = create the project's secrets folder (once). set "Jwt:Key" "..." = save a secret. list = show them.
[NEW] REST Client / .http file	A VS Code extension for sending requests from a text file. Click Send Request (not the URL, which opens the browser with GET).
[NEW] # @name / {{login.response.body.token}}	REST Client variables: name a request, then reuse a field from its answer, like the token.
[NEW] F12 -> Network	The browser's developer tools tab that shows each request and its status code.